

Spreading Out Across Expanding Idea Space

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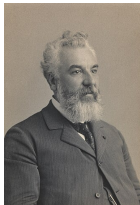
University of Essex

September 2026

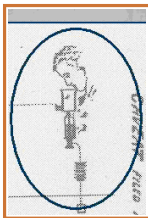
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February 14, 1876: A Collision in Idea Space



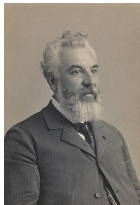
Alexander Graham Bell



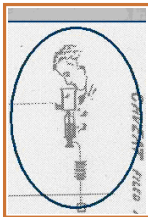
Elisha Gray

A **patent interference** — two inventors, same invention

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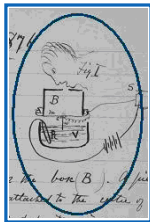
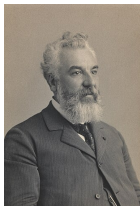


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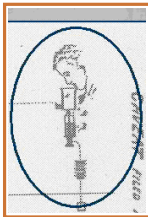


If every invention a
“location” in idea space

February 14, 1876: A Collision in Idea Space



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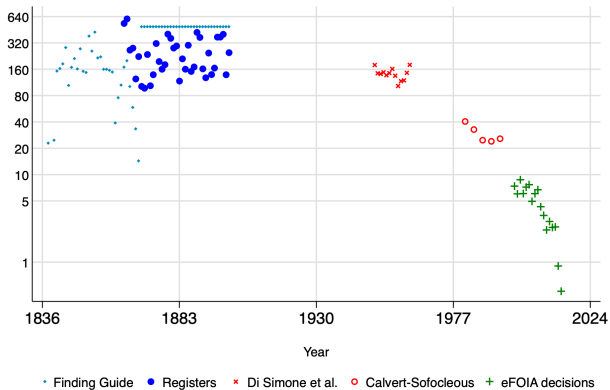
A **patent interference** — two inventors, **same invention** $\longleftrightarrow$ They chose the **same location** in idea space



Such Collisions Were Once Routine

≈ 1 in 20 patents (1870s) $\rightarrow <1$ in 10,000 (2014)!

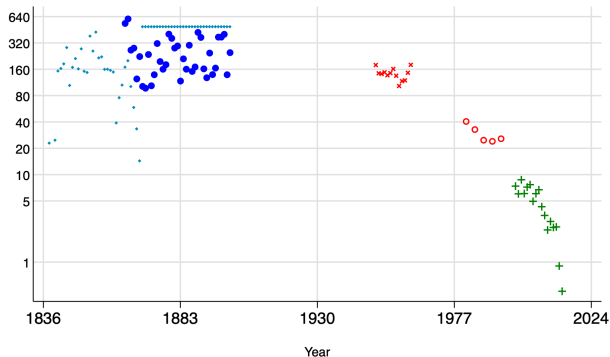
Interference rate per 10,000



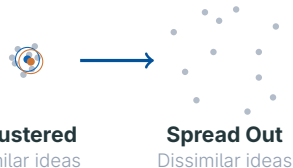
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• Finding Guide • Registers • Di Simone et al. • Calvert-Sofocleous • eFOIA decisions

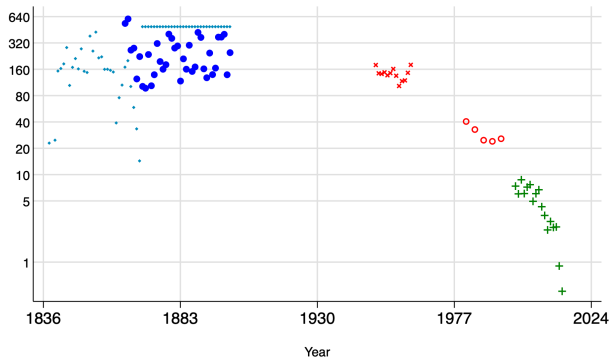


Our claim: inventors spread out across an expanding idea space

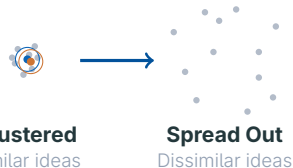
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Our claim: inventors spread out across an expanding idea space & **where inventors stand matters for growth.**

Three Questions, Three Routes

Questions

1. Are inventions spreading out?
2. Why are they spreading out?
3. What are the consequences?

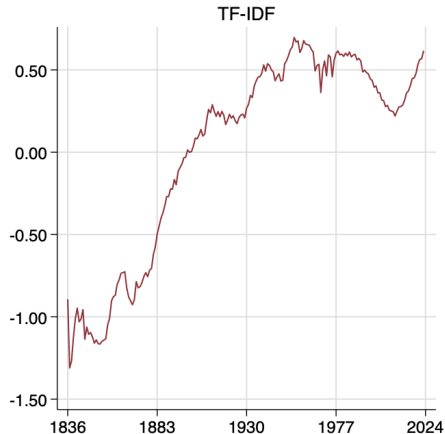
How we answer

1. **Build & validate map** of ideas
2. Develop **theory** of spreading
3. **Test** and quantify consequences

Are Inventions Spreading Out?

Let's Check with Text

Standardized avg pairwise cosine similarity by year

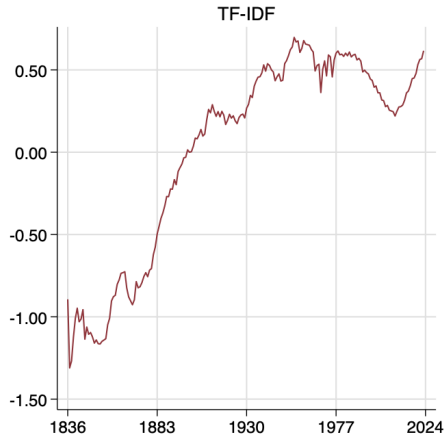


Data: Claims text of 11M+ US patents

Standard Tool: **TF-IDF** word frequencies

Let's Check with Text

Standardized avg pairwise cosine similarity by year



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Similarity is rising?

✗ This says inventions are *crowding* —
contra the interference record.

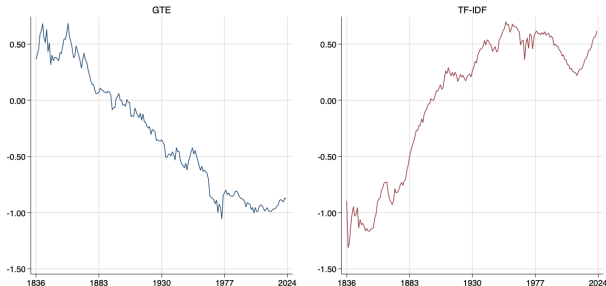
Same Text, Opposite Maps

Seven representations:

TF-IDF · GTE · PaECTER · S-BERT
Doc2Vec · USE · OpenAI

Divergent conclusions:

- TF-IDF: similarity *rising*
- Others: similarity *falling*



Which “map” should we trust?

Three Benchmarks Test Technological Similarity

A useful map recovers similarity across **judges**, **scales**, and **eras**.

Judges and benchmarks	Similarity scale	Coverage in time
Same-invention collisions Examiner-declared interferences	Fine: the same patentable invention	2001–2014 
Human similarity rankings Lay judgments	Relative: which pair is more similar?	1850–1975 
Shared CPC categories USPTO classifications	Coarse: shared section or class	1850–2023 

Validation Selects GTE Embeddings

		Same-invention collisions	Human rankings	Shared CPC categories	
		F10	Agreement	Section ROC	Class ROC
✓	GTE	0.90	0.62	0.596	0.656
	PaECTER	0.90	0.51	0.590	0.672
	S-BERT	0.82	0.54	0.600	0.671
×	TF-IDF	0.77	0.35	0.514	0.525

Higher is better in every column

✓ Select GTE

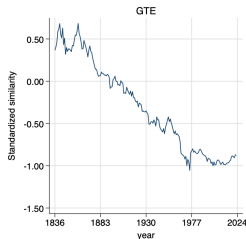
Strongest historical agreement
Nearly tied on same-invention recovery
Competitive on CPC boundaries

× Reject TF-IDF

Beats chance, but ranks last on every benchmark and **produces the wrong map**

► bicycle ≠ velocipede

Three Validated Maps Agree: Inventions Are Spreading Out



GTE shown;
PaECTER and S-BERT
exhibit the same secular
decline

Best map: $\approx 1.0\text{--}1.5\sigma$ decline in similarity, 1838–2023 ^{GTE}
All 3 validated maps: $\approx 0.8\text{--}1.0\sigma$ decline, 1900–2000

Spreading out:

- ✓ after 2000 among **independent inventions**
- ✓ at **local and global spatial scales**
- ✓ **within and between** CPC classes
- ✓ **within classes as they age** and **across cohorts**

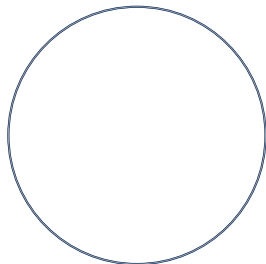
Complementary evidence:

- **Interference rates declined**
- **Embedding space expanded**

Why Are Inventions Spreading Out?

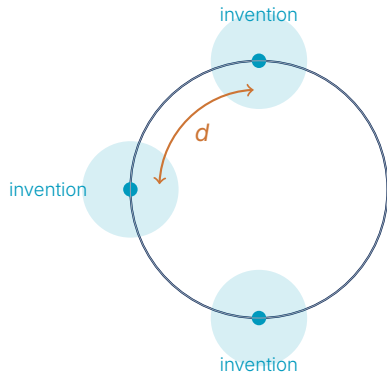
Spacing Gives Inventors Market Power

- **Idea space:** Circle of circumference H — similar problems, similar solutions



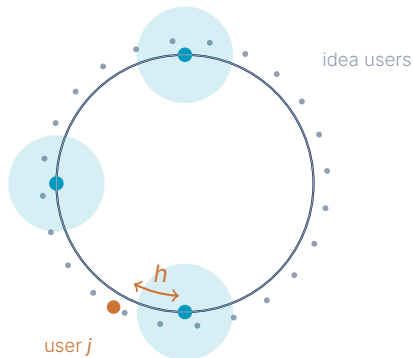
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- **Idea space:** Circle of circumference H — similar problems, similar solutions
- **Idea producers:** Sell non-exclusive license to use non-rival idea. Choose **entry** f & **position**; **quality** q_i & **price** p_i .
Maximize profit $\pi_i = R_i - c(q_i) - f$



Spacing Gives Inventors Market Power

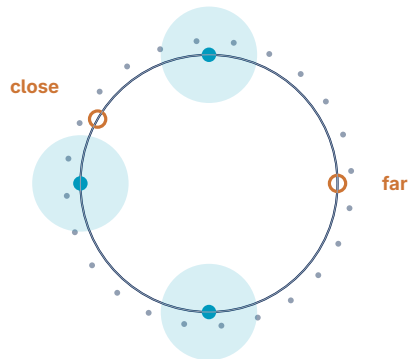
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- **Idea users:** Distributed uniformly on circle. Choose **one invention** to maximize **net log TFP gain** $A_j - p_i \equiv Q_i - \tau h - p_i$



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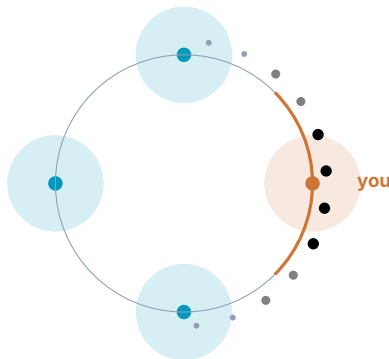
Distance reduces close substitutes.



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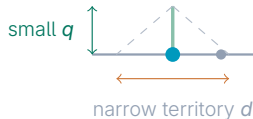
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Distance reduces close substitutes.
Isolation widens your market.



A Larger Territory Requires a Larger Invention

Close to neighboring inventor

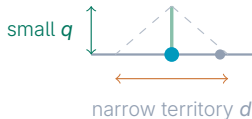


In symmetric equilibrium,

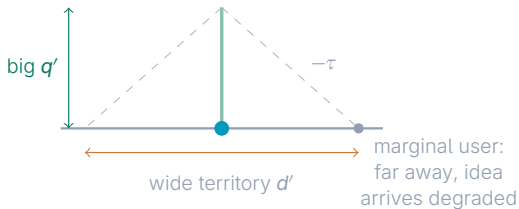
- Your idea users sit up to $d/2$ away.

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Far from neighboring inventor



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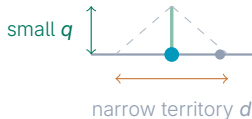
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The **farther** your marginal user:

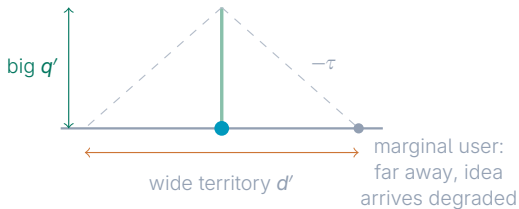
- **More pricing power**
(users have no nearby alternatives)
- **More quality needed**
(to be worth paying adaptation costs)

A Larger Territory Requires a Larger Invention

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In symmetric equilibrium,

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The **farther** your marginal user:

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(users have no nearby alternatives)
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(to be worth paying adaptation costs)

Spacing pins quality exactly.

$$q^* = d/\gamma$$

Equilibrium Turns Spacing into Prices, Quality, and Entry

Geometry pins **where you stand**, **what you charge**, and **how big you build**.

1. Isolation sets your markup:

$$p^* = \tau d$$



► Details

► Existence

► $\tau\gamma > 1/2$

► Comp statics

► Eqns

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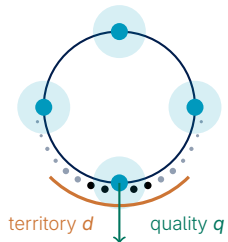
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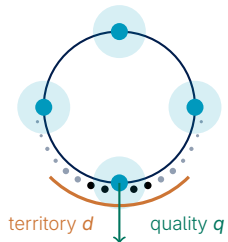
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3. Territory revenue must cover R&D & entry:

$$d^* = \sqrt{\frac{f}{\tau - \frac{1}{2\gamma}}}$$

(free entry, zero profit)



free entry: $pd - c(q) = f$

► Details

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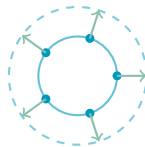
A Growing Frontier Raises Entry Costs and Spreads Inventors Out

As innovation expands frontier... Weitzman 1998

$$\dot{H} = \delta \cdot n \cdot (Q - \tau d/4)$$

...the burden of knowledge grows... Jones 2009

$$f = \phi H^\alpha \quad \text{baseline } \alpha = 1$$



Only rising H, f deliver spreading out & expanding variety simultaneously

► KPF

► Scenarios

► α

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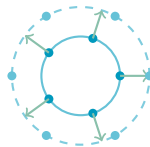
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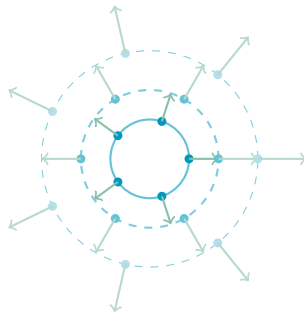
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The Ripple:

$$H \uparrow \Rightarrow f \uparrow \Rightarrow d \uparrow n \uparrow q \uparrow \Rightarrow H \uparrow \Rightarrow \dots$$

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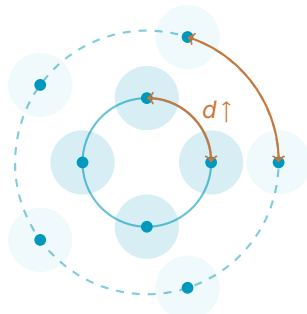
The Ripple's Fade: Attenuating Knowledge Spillovers

Gross quality includes direct diffusion from local spillovers:

$$Q_i = \underbrace{q_i}_{\text{purchased idea}} + \underbrace{\beta\left(1 - \frac{d}{\lambda}\right)}_{\substack{\text{spillovers} \\ \text{(declines with } d\text{)}}} q$$

An endogenous brake on growth

Fast growth early, when inventions are packed; deceleration as they spread.



halos fade: $\beta(1 - d/\lambda) \rightarrow 0$

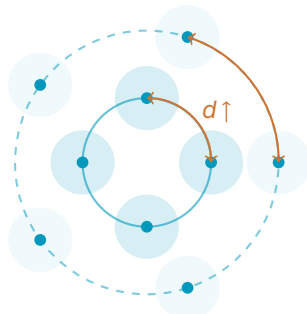
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Inventors spread out, build bigger, and borrow less. cf. Fort et al. 2026

Declining R&D Productivity From Local Ideas That Are Costly to Scale

**R&D
productivity**

$$\Pi \equiv \frac{g_{TFP}}{\text{Agg R\&D}}$$

cf. Bloom et al. 2020

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TFP gains are local

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g_{TFP}



cf. Bloom et al. 2020

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R&D summed over idea space

$$\text{Agg R\&D} = \underbrace{\boxed{n}}_{\text{inventions}} \cdot \underbrace{\left[\frac{1}{2} \gamma q^2 + \phi H \right]}_{\text{variable} + \text{fixed costs}}$$

$n \times \text{costs}$



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Ideas useful only locally $\Rightarrow$ TFP growth is **average** local gain
— while R&D is **sum** across an expanding idea space $\Rightarrow$ Declining Π
(alongside **weakening spillovers** & traditional **fishing out and burden of knowledge** channels)

Growth Depends on the Distance Between Inventions

Frontier growth pinned by **cost and spillover parameters** and spacing:

$$g_H = \underbrace{\delta \left(\frac{1}{\gamma} - \frac{\tau}{4} \right)}_{\text{R\&D and adaptation costs}} + \underbrace{\frac{\delta \beta}{\gamma} \left(1 - \frac{d}{\lambda} \right)}_{\text{spillovers attenuate as inventors spread out}}$$

- Cheaper entry, R&D, adaptation, and stronger spillovers all raise growth
- **Spatial implication:** **where inventors stand** enters the growth rate

► Detailed equations

► General model

► Equations

The Arrangement of Ideas Shapes Growth

Many growth models track a scalar stock of knowledge.

Here, growth also depends on how far apart inventions stand.

Effective spillover intensity

Romer 1990

$$\beta(1 - d^*/\lambda)$$

Innovation step size

Aghion-Howitt 1992

$$q^* = d^*/\gamma$$

Research productivity

Jones 1995

$$\Pi = g_{TFP} / \text{Agg R\&D}$$

Research scale

Jones 1995

$$n^* = H/d^*$$

What Are the Consequences?

One Geometry Unifies Many Facts

Spreading out (d^* , n^* $\uparrow$):

✓ Patent similarity falling

this paper, Chiopris 2024

✓ Spillovers attenuating

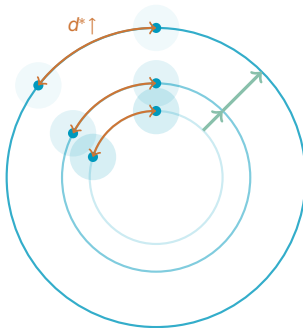
Lucking+ 19, Fort+ 26, Berkes-Gaetani 26

✓ R&D productivity declining

Jones 95, Kortum 97, Bloom+ 20

✓ Varieties increasing

Patent counts, Hirschey+ 12



► Comparative statics

► Kelly replication

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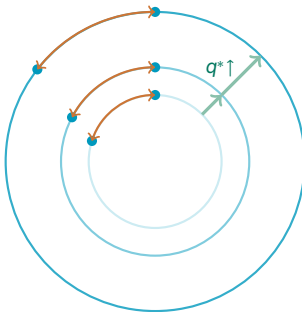
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Building bigger (q^* , p^* $\uparrow$):

✓ R&D increasing

Hirschey+ 12

✓ Breakthrough rates rising

Kelly+ 2021 (horiz+vert)

✓ Patent values rising

Hall+ 05, Bessen+ 18, Kogan+ 17

Spreading out within fields reconciles Howitt/Peretto with Bloom+2020 evidence. Attenuating spillovers provide structural foundation for Fort+2026. Spacing and quality together reinterpret breakthrough patents (Kelly+2021).

► Comparative statics

► Kelly replication

The Model's Fingerprint: Spacing Couples with Quality

This model's distinctive prediction:

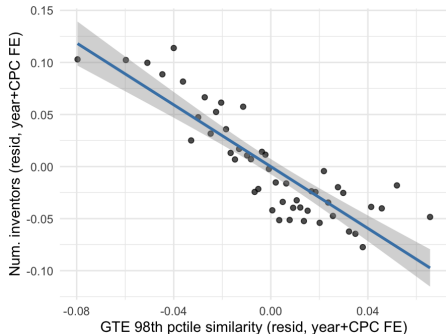
$$q^* = d/\gamma$$

More isolated inventions:

- Use **larger teams**
- More often **firm-assigned**
- More often **cited**
- Have **higher market value**

(KPSS: $\approx \$22\text{--}33\text{M}$ per σ of isolation)

Holds in cross-section (220K patents, within class-year)
and across the **188-year** class panel (CUSP).



Co-inventors vs. similarity
residualized on year $\times$ CPC class

► All four proxies

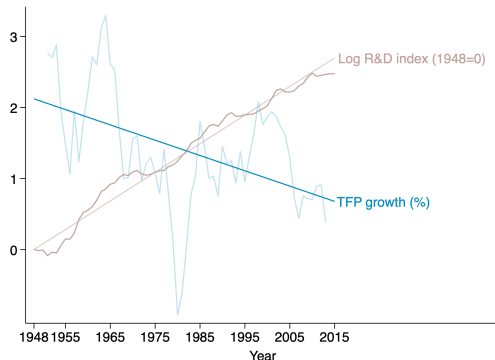
► Regression table

Research Productivity Falls Through Two Margins

$$g_{\Pi} = \underbrace{g_{g_{TFP}}}_{-1.6\%/yr} - \underbrace{g_{R\&D}}_{+4.0\%/yr} = \boxed{-5.6\%/yr}$$

How much of the **TFP-growth slowdown** comes from spreading?

How much of **R&D growth** comes from spatial forces?



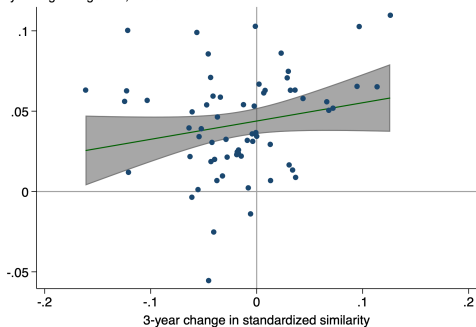
Bloom, Jones, Van Reenen and Webb, 2020

Strategy: Quantify the spatial contribution to each margin, then add them.

The Data Supply Two Moments

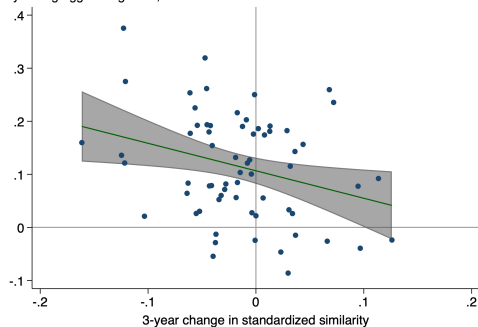
TFP margin

3-year log TFP growth, 1949-2015



R&D margin

3-year log agg R&D growth, 1949-2015



More spreading $\iff$ slower TFP growth

More spreading $\iff$ faster aggregate R&D

Spreading growth: $g_d \approx -\Delta \overline{\text{Sim}}_t$

Next: the model interprets these comovements

The Model Gives Both Moments Their Structural Meaning

TFP

1. Model → Regression

Spillover drag from spreading depends on spacing.

$$\Delta \log(\text{TFP})_t = b_0 + b_1(-\Delta \overline{\text{Sim}}_t) \\ + b_2(-\Delta \overline{\text{Sim}}_t)(-\overline{\text{Sim}}_{t-1}) + b_3 t + \epsilon_t$$

2. Estimates

Long-run spillover drag: **-0.11 pp/yr**

1948–2015; enters the decomposition

1981–2001 benchmark: -0.16 pp/yr

The Model Gives Both Moments Their Structural Meaning

TFP

1. Model → Regression

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2. Estimates

Long-run spillover drag: **-0.11 pp/yr**

1948–2015; enters the decomposition

1981–2001 benchmark: -0.16 pp/yr

R&D

1. Model → Regression

$$\Delta \log(\text{R\&D})_t = a_0 + a_1(-\Delta \overline{\text{Sim}}_t) + \epsilon_t$$

2. Estimates

Slope $a_1 \Rightarrow \hat{\theta} = 72\%$ Variable-cost share

Quality governs the spacing response

Intercept $a_0 \Rightarrow \hat{g}_H = 2.67\%/yr$ Frontier growth

Variety and entry costs govern trend

► TFP bridge

► R&D bridge

Four Quantitative Checks Support the Decomposition

Check	Our Estimate	Benchmark
<i>External validation</i>		
Spillover drag validates TFP regression	-0.16 pp/yr TFP regression	-0.15 pp/yr ✓ Bloom+2013 spillover elasticity $\times$ our spreading: $0.206\text{pp} \times 1.04/\sigma \times -0.007\sigma/\text{yr}$
Variable-cost share validates R&D regression	$\hat{\theta} = 72\%$ R&D regression	69% ✓ NSF BERD R&D labor share
<i>Internal consistency</i>		
Idea - space growth overidentified: two estimates agree	$\hat{g}_H = 2.67\%/\text{yr}$	$g_H = 2.7\%/\text{yr}$ ✓ implied by $g_{R\&D} = \frac{3}{2} g_H$
Spacing vs. R&D growth model ratio holds in data	$g_d/g_{R\&D} = 1/3$ model	$0.2-0.4$ ✓ data

Spatial Forces Account for 42% in the Baseline

TFP regression

0.11 pp/yr

Long-run spatial drag
1948–2015

Model partition of observed R&D growth: 4.0%/yr

Components in percentage points per year

Entry expansion $(1 - \frac{\alpha}{2})g_H$	1.33
Quality scaling $(\theta \frac{\alpha}{2})g_H$	0.92
Spatial subtotal	2.25
Fishing out $(\theta \eta \frac{\alpha}{2})g_H$	0.92
Burden of knowledge $(1 - \theta)\alpha g_H$	0.83

Add the spatial contributions; divide by the observed 5.6%/yr decline.

$$\frac{\overbrace{0.11}^{\text{TFP}} + \overbrace{2.25}^{\text{R\&D}}}{5.6} = \boxed{42\%}$$

Sensitivity: 42–58%

► Detailed accounting

► Sensitivity

Invention spacing shapes competition and spillovers.

Tools from spatial economics

- Distance costs
- Local spillovers
- Location choice and market areas

Their role in idea space

- Users adapt distant ideas
- Nearby ideas share knowledge
- Inventors choose projects

Spatial tools may be first - order for understanding growth.

Three Takeaways

1

Inventions spread out

Validated NLP measures spanning 1836–2023; Interferences 1870–2014.

2

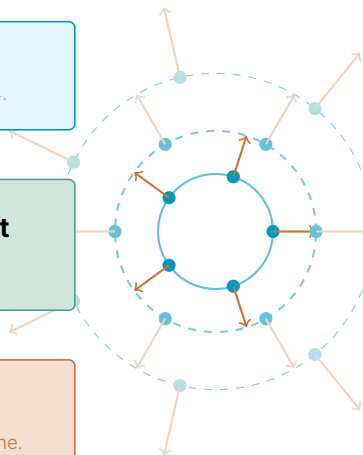
An expanding idea frontier drives inventions apart

Unifies empirical patterns; Endogenizes core growth primitives.

3

Where inventors stand matters for growth

Spatial forces account for 40–60% of the long-run R&D productivity decline.



Backup Slides

Backup: Interference Data

◀ Return

NAME OF PARTY	DATE	NAME OF BLAINE	DATE
Ehrlich, Leo Lawton, Jas B - (1873)	Jan 13 1890	Roll Paper Cutters Statement of Lawton Dec. 22, 1889 Statement of Ehrlich Jan 6 & 1890	Decided in favor Lawton, May 1890 D. R. May 1890 Rehearsal May 1890
Blaine, David H Hadley, William R - (1874)	Apr 20 1890	Corn Harvesters Statement of Blaine Apr 20, 1890 Brief for Hadley Apr 21, 1890 Statement of Hadley Apr 25, 1890 Statement of Blaine May 7 & 1890 Motion of Hadley for leave to amend his paper May 6, 1890 Brief for Hadley May 6, 1890 Rehearsal of Blaine Hadley May 6, 1890	Decided in favor Hadley, Oct 1890 D. R. May 1890 Rehearsal May 1890

Example (Jan 7, 1890)

Ehrlich v. Lawton: Roll paper cutters

Blaine v. Hadley: Corn harvesters

Purpose - digitized from National Archives:

- USPTO *Registers of Interferences*, 1864–1900
- 19,388 interference cases documented
- Average 504/year (2.7–4.9%)

Other sources:

- Case Files (1838–1900): 87–880/year
- DiSimone et al. (1950–1962): 640/year (1.4%)
- Calvert & Sofocleous (1980–1994): 237/yr (0.3%)
- Ganguli et al. (1998–2014): 76/year (0.05%)

Historical (1836–1975): ProQuest Patents Core

- OCR-digitized patent images
- Full text of claims extracted
- Quality varies with original document condition

Modern (1976–2023): USPTO PatentsView

- Machine-readable full text
- Structured data with claim parsing
- Consistent quality

Potential discontinuity at 1976:

- Some evidence of break in levels
- Trends consistent across periods
- Results robust to excluding transition years

Backup: Computing Similarity Efficiently

[◀ Return](#)

Challenge: $O(N^2)$ pairwise comparisons infeasible for millions of patents

Solution: For unit-normalized vectors, average cosine similarity reduces to:

$$\bar{S} = \frac{1}{N(N-1)} \sum_{i \neq j} \cos(v_i, v_j) = \frac{\|\sum_i v_i\|^2 - N}{N(N-1)}$$

Complexity: $O(N \cdot d)$ where d = embedding dimension

Implementation:

1. Normalize all vectors to unit length
2. Sum vectors: $S = \sum_i v_i$
3. Compute $\|S\|^2$
4. Apply formula

Cross-sectional SD: Subsample up to 10,000 patents/year

Main specification: Standardize by annual cross-sectional SD

Robustness checks:

1. Time-invariant global SD → Nearly identical results
2. Raw similarity (no standardization) → Same qualitative pattern
3. Different sample sizes per year → Robust

Why standardize?

- Different representations have different scales
- No intrinsic economic interpretation of raw similarity
- SD provides meaningful units for comparison

Backup: Interference Validation Task

[◀ Return](#)

Patent interferences (2001–2014):

- **First to invent:** USPTO proceeding for multiple applicants w/ identical claims
- Provides ground truth for “identical” similarity
- 322 true interfering pairs among 96,580 application pairs

Economic intuition: Examiner ranks pairs by similarity, investigates above threshold

- Higher threshold → fewer false positives but miss true interferences
- Lower threshold → catch more but burden staff with unnecessary investigations

Metrics:

- F10: Weights recall 10× more than precision (missing interferences is costly)
- PR AUC: Precision-Recall area under curve across all thresholds

Key result: GTE, PaECTER, OpenAI retrieve ~90% of true interferences with 2–5× fewer false positives than TF-IDF/S-BERT

Backup: Human Annotation Task

[◀ Return](#)

Historical patents (1850–1975):

- Sampled patent pairs that each model ranked at least 50 percentiles apart
- Annotators rank **relative** similarity of 2 patent pairs
- Tests temporal robustness (historical language): Oversample 1880–1920

Task: Do model rankings agree with human rankings?

- For each patent, rank others by similarity
- Compare model ranking to human ranking

Metric: Agreement coefficient from regression

$$\text{Human Rank} = \alpha + \beta \cdot \text{Model Rank} + \epsilon$$

Higher β = better agreement

Backup: Classification Validation Task

[◀ Return](#)

USPTO Classifications (1850–2023):

- CPC technology codes assigned by examiners
- Section level (8 categories) and Class level (120+ categories)
- Captures expert judgment of technological relatedness

Task: Predict whether patent pair shares classification

- Same Section (coarse): 8 top-level categories
- Same Class (fine): 3-digit classification

Metric: ROC AUC

- Area under Receiver Operating Characteristic curve
- 0.5 = random, 1.0 = perfect

Backup: Why TF-IDF Fails

[◀ Return](#)

TF-IDF overweights period-specific language:

- Treats “velocipede” (1880s) and “bicycle” (modern) as unrelated
- Period-specific terminology dominates similarity scores
- Creates spurious correlation with time

Example: 1880 velocipede patent

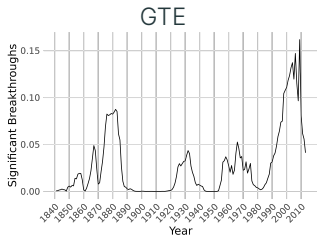
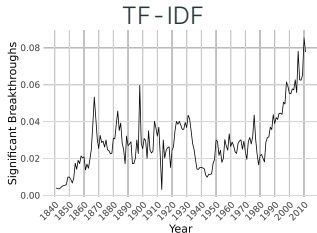
- TF-IDF: High similarity to other 1880s patents (shared vocabulary)
- GTE: High similarity to modern bicycle patents (shared concepts)

Evidence:

- TF-IDF similarity correlates with word overlap
- GTE similarity correlates with conceptual similarity
- Google Ngrams shows vocabulary shifts over time

Backup: Kelly et al. Breakthrough Replication

◀ Return



Kelly et al. (2021): Identify “breakthrough” patents using similarity to future patents

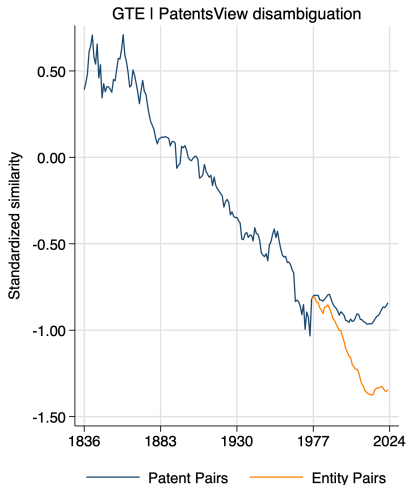
Our replication with GTE:

- Qualitative conclusions align (more breakthroughs today)
- Quantitative results more robust (less sensitivity to methodological choices)
- TF-IDF produces noisier breakthrough classification

Implication: Validated similarity measures improve downstream analyses

Backup: Robustness — Multi-Patent Entities

◀ Return



Concern: Post-2000 dynamics coincide with: business method patents, non-practicing entities, increased defensive patenting.

▶ Patents v entities

- **Multiple patents from same entity may be similar but not independent.**

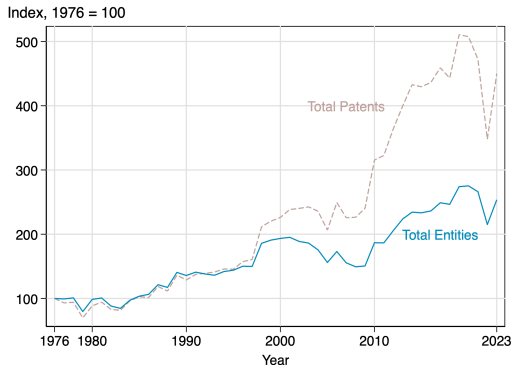
Strategy: Sample 1 patent/entity-year

Result:

- Decline persists after correction
- **Independent inventions** still spreading out

Backup: Growth in Patents vs. Patenting Entities

[Return](#)



- Number of issued utility patents and unique patenting entities per year
- Divergence after 1999: substantial growth in patents per entity
- Driven by business method patents and non-practicing entities
- Motivates sampling 1 patent per entity per year for robustness

Backup: Robustness — Spreading Out Within Technology Classes

Alternative explanations: Changing patent office practice over time? Shifts across major technology areas?

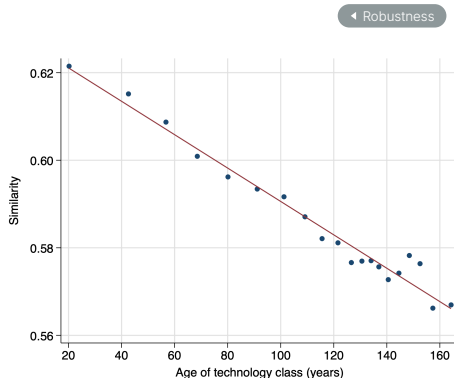
Test: Within-class similarity by class “age”

- Birth = Class first issued 50 patents
- e.g., Combinatorial Chemistry 2001
- Addresses compositional concerns

► Between

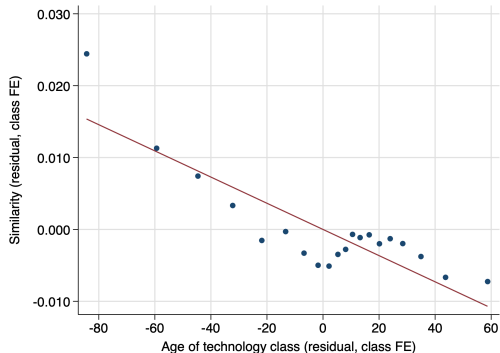
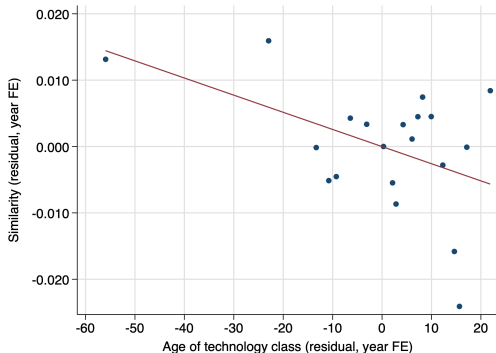
Finding: Within-class similarity declines as classes mature

► Class+Year FE



Spreading out is a dynamic process tied to field evolution

Backup: Similarity Within Class by Class Age

[Return](#)

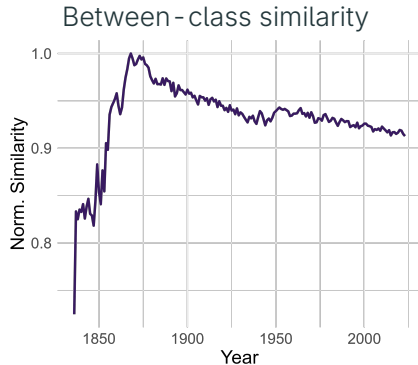
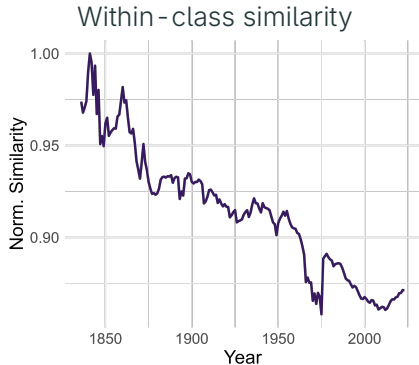
Right: Cross-cohort comparison (year FE) — robustness to patent office practice

Left: Within-cohort comparison (class FE) — robustness to composition

Backup: Within vs. Between Technology Classes

◀ Within classes

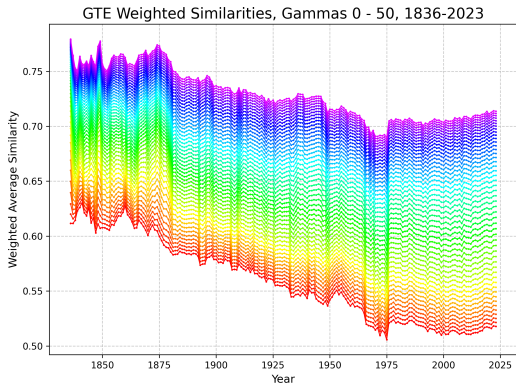
◀ Robustness



- **Addresses compositional concern:** Decline not driven by shifts across technology fields — spreading out occurs *within* established classes

Backup: Similarity at Different Spatial Scales

◀ Robustness

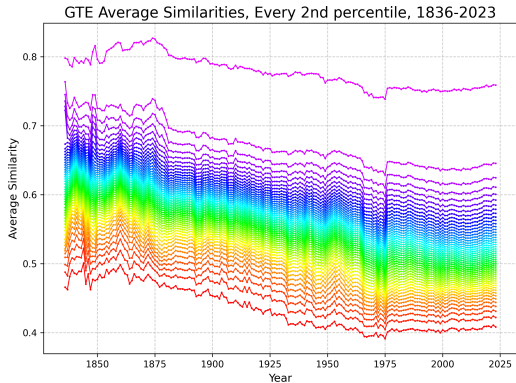


Weighted average: $\equiv \frac{1}{n} \sum_{i=1}^n \frac{\sum_{j \neq i} (1-d_{ij}) e^{-\gamma d_{ij}}}{\sum_{j \neq i} e^{-\gamma d_{ij}}}$
where γ from 0 (global) to 50 (local)

- **Key finding:** Similar declining trends across all spatial scales
- Model predictions concern averages — important to verify pattern holds across distribution
- Post-2000 arrest slightly stronger at local scales (consistent with entity correction)

Backup: Similarity at Different Quantiles

◀ Robustness



Similarity at Different Quantiles:

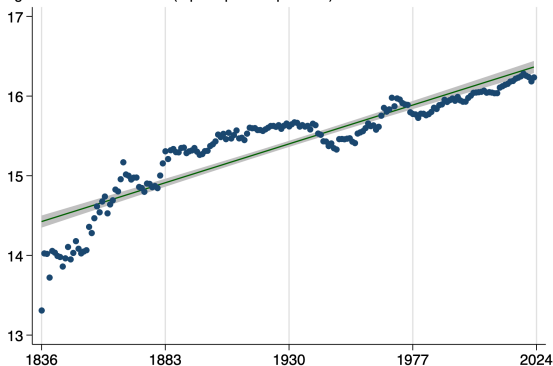
- 50 quantiles of pairwise similarity in each year
- Secular decline is robust across all quantiles
- Post-2000 increase in similarity is slightly faster for higher quantiles

Corroboration: Expanding Convex Hull

◀ Robustness

◀ R&D regression

Log volume of convex hull (7 principal components)



Is Idea Space Expanding?

Test:

- 1024 GTE dimensions to 7 PC
- Measure volume of convex hull

Result:

- +0.5%/yr (6 PC: +0.4%/yr)
- Likely under-estimate due to dimensionality reduction

Why linear in log TFP? $A_i(h) = Q_i - \tau h$

Standard in spatial competition (Salop 1979):

- Idea consumers have preferences linear in quality net of distance costs
- $A_i(h)$ interpreted as log TFP $\Rightarrow$ firms care about *proportional* productivity gains

Microfoundation: Each downstream firm has one unit of fixed input ℓ and produces:

$$y = e^A \cdot \ell$$

With output price = 1 and $\ell = 1$, profit is $\pi = e^A$. Willingness to pay for technology delivering incremental log TFP A (relative to baseline $e^0 = 1$):

$$WTP = e^A - 1 \approx A \quad (\text{first-order Taylor approximation})$$

Accuracy: For annual TFP increments ($A \approx 0.015/\text{year}$), approximation error $< 0.01\%$

Advantage: Predictions directly comparable to empirical TFP elasticities (Bloom et al. 2013) and growth accounting (Bloom et al. 2020)

Backup: Model Equations

◀ Equilibrium

◀ Growth rates

Fixed cost (burden of knowledge):	$f(H) = \phi H^\alpha$
R&D cost:	$c(q_i) = \frac{1}{2}\gamma q_i^{1+\eta}$
Realized quality (with spillovers):	$Q_i = q_i + \frac{1}{2}\beta\left(1 - \frac{d}{\lambda}\right)(q_{i-1} + q_{i+1})$

Equilibrium pricing and quality:	$p^* = \tau d, \quad q^* = \frac{d}{\gamma}$
Equilibrium spacing:	$d^*(H) = \sqrt{\frac{\phi H}{\tau - \frac{1}{2\gamma}}}$
Equilibrium entry:	$n^* = \frac{H}{d^*}$
Equilibrium revenue:	$R^* = p^* \cdot d^* = \tau d^2$

Growth rates (baseline $\alpha = 1, \eta = 1$):	
Spacing, quality, entry:	$g_d = g_q = g_n = \frac{1}{2}g_H$
Fixed cost:	$g_f = g_H$
Aggregate R&D:	$g_{R\&D} = \frac{3}{2}g_H$

Backup: Equilibrium Existence Conditions

Spreading - out condition: ▶ Proposition

◀ Return

$$\tau\gamma > \frac{1}{2}$$

Marginal revenue of expanding territory exceeds marginal cost.

Second - order conditions:

- Pricing: $\partial^2 R / \partial p^2 < 0$ (satisfied)
- Quality: $\partial^2 \pi / \partial q^2 = -\gamma < 0$ (satisfied)
- No spatial deviation (verified in paper)

Additional conditions:

- Spillover reach: $d < \lambda$ (spillovers active)
- Full coverage: All downstream firms adopt some technology

Equilibrium: Pricing and Quality

[◀ Return](#)

Symmetric equilibrium: n inventors, equal spacing $d = H/n$, identical (p, q)

1. Pricing. Boundary firm at $\tilde{h}$ indifferent between inventor i and neighbor:

$$Q - p_i - \tau \tilde{h} = Q - p - \tau(d - \tilde{h}) \Rightarrow \tilde{h} = \frac{d}{2} + \frac{p - p_i}{2\tau}$$

Revenue $R_i = 2p_i \tilde{h}$; FOC $\partial R_i / \partial p_i = 0$ yields: $p^* = \tau d$

2. Quality. MR = MC for quality investment ($\partial Q_i / \partial q_i = 1$, $\partial \tilde{h} / \partial Q_i = 1/2\tau$):

$$\frac{p_i}{\tau} = \gamma q_i \Rightarrow q^* = \frac{d}{\gamma} \quad (\eta = 1)$$

3. Zero-profit condition. Substitute $R = \tau d^2$, $c(q^*) = \frac{d^2}{2\gamma}$:

$$\tau d^2 - \frac{d^2}{2\gamma} - f = 0 \Rightarrow d^* = \sqrt{\frac{f}{\tau - \frac{1}{2\gamma}}}$$

Backup: Free Entry Determines Equilibrium Spacing and Inventions

[◀ Return](#)

Zero-profit condition:

$$\underbrace{\tau d^2}_{\text{Revenue}} - \underbrace{\frac{d^2}{2\gamma}}_{\text{R\&D cost}} - \underbrace{f}_{\text{Entry cost}} = 0$$

Solving for equilibrium spacing and number of inventions ($n = H/d$):

$$\boxed{d^* = \sqrt{\frac{f}{\tau - \frac{1}{2\gamma}}}} \Rightarrow \boxed{n^* = H \sqrt{\frac{\tau - \frac{1}{2\gamma}}{f}}}$$

Symmetric equilibrium p^*, q^*, d^*, n^* in terms of costs τ, γ, f , and market size H

Backup: Spreading Out — Proposition

◀ Return

Proposition (Spreading Out)

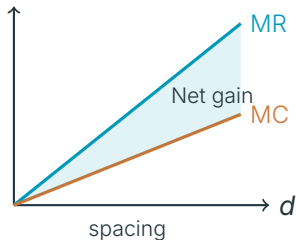
For $\tau\gamma > \frac{1}{2}$, equilibrium spacing increases with opportunity space: $\frac{dd^*}{dH} > 0$.
Inventions become **less similar** over time.

Adaptation costs must create sufficient pricing power

▶ Intuition

Backup: Spreading Out — Why Is It Profitable?

Why is spreading out profitable?



Marginal revenue of expanding territory:

- Revenue $R = \tau d^2 \Rightarrow MR = 2\tau d$

Marginal cost of expanding territory:

- Need higher quality: $q = d/\gamma$
- $MC = d/\gamma$

Spreading profitable when:

$$MR > MC \Rightarrow \boxed{\tau\gamma > \frac{1}{2}}$$

Adaptation costs must create sufficient pricing power

◀ Proposition

Backup: Comparative Statics Derivations

Spreading out: From zero-profit condition $d^2(\tau - \frac{1}{2\gamma}) = \phi H$:

◀ Equilibrium

◀ Literature Evidence

$$\frac{dd}{dH} = \frac{\phi}{2d(\tau - \frac{1}{2\gamma})} = \frac{\phi}{dR/dd - dc/dd} > 0$$

Rising quality and prices:

$$\frac{dq}{dH} = \frac{1}{\gamma} \frac{dd}{dH} > 0, \quad \frac{dp}{dH} = \tau \frac{dd}{dH} > 0$$

Rising entry:

$$\frac{dn}{dH} = \frac{1}{d} - \frac{H}{d^2} \frac{dd}{dH} > 0 \text{ under spreading-out condition}$$

Declining R&D productivity: See ▶ Force decomposition

Backup: Robustness to Entry Cost Curvature

[◀ Return](#)

With $f(H) = \phi H^\alpha$, equilibrium spacing satisfies $d \propto H^{\alpha/2}$:

Condition	Entry Growth	Prediction
$0 < \alpha < 2$	$g_n = (1 - \frac{\alpha}{2})g_H > 0$	Entry grows ✓
$\alpha = 2$	$g_n = 0$	Entry stagnates
$\alpha > 2$	$g_n < 0$	Entry declines ×

Main results robust for $\alpha < 2$:

- Spreading out: $g_d = \frac{\alpha}{2}g_H > 0$ for any $\alpha > 0$
- Declining R&D productivity: Holds throughout range
- Higher $\alpha \rightarrow$ faster spreading, but lower spatial share of productivity decline

Counterfactual boundary: Patent counts grow over time, ruling out $\alpha \geq 2$

Backup: How Entry Costs Shape Equilibrium

[◀ Return](#)

As idea space H grows, what happens to d^*, n^*, q^*, p^* ?

Scenario	d^*, q^*, p^*	Varieties n^*
1. f constant	unchanged	$\uparrow$ (linear in H)
2. $f(H)$ decreasing	$\downarrow$ (clustering!)	$\uparrow\uparrow$ (faster)
3. $f(H)$ increasing (KPF baseline)	$\uparrow$ (spreading)	$\uparrow$ (grows with H)
4. $f(H)$ increasing rapidly	$\uparrow\uparrow$ rapidly	$\downarrow$ (fewer)

Only rising H, f is consistent with spreading out and expanding varieties

Backup: A Knowledge Production Function

◀ Return

Evidence of expansion:

- Patent filings: $\approx 500/\text{yr}$ (1840s) $\rightarrow$ 350,000/yr (2020s)
- Each discovery opens **adjacent research directions** Weitzman 1998; Scotchmer 1991
- Electricity $\rightarrow$ electronics $\rightarrow$ computing $\rightarrow$ AI...

Formalizing the intuition:

$$\dot{H} = \delta \cdot n \cdot \underbrace{(Q - \tau d/4)}_{\text{net quality contribution}}$$

- δ : efficiency of knowledge in opening new directions
- n inventions, each contributing to frontier expansion

Key features:

- Uses *social* quality Q (incl. spillovers), not private effort q
- Net of adaptation costs—only “usable” quality opens new directions
- Standard supply-side KPF (Romer 1990; Jones 1995; Kortum 1997)

Backup: TFP and R&D Growth Equations

[◀ Growth rates](#)[◀ TFP regression](#)[◀ R&D regression](#)[◀ Decomposition](#)

TFP growth:

$$g_{TFP} = \underbrace{g_q \left(1 + \beta - \frac{\beta d}{\lambda} \right)}_{\text{Quality (with spillovers)}} - \underbrace{\frac{\beta q}{\lambda} g_d}_{\text{Spillover attenuation}} - \underbrace{\frac{\tau}{4} g_d}_{\text{Adaptation drag}}$$

R&D growth:

$$g_{R\&D} = \underbrace{g_n}_{\text{Entry}} + \underbrace{\theta \cdot g_q}_{\text{Quality scaling}} + \underbrace{\theta \cdot g_q}_{\text{Fishing out}} + \underbrace{(1 - \theta) g_f}_{\text{Burden of knowledge}}$$

where θ = variable cost share; $\alpha = 1$, $\eta = 1$

Backup: Growth Equations with General Cost Curvatures

[◀ Growth Rates](#)[◀ Decomposition](#)

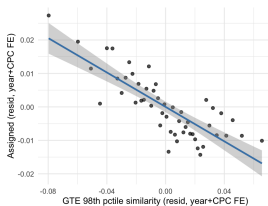
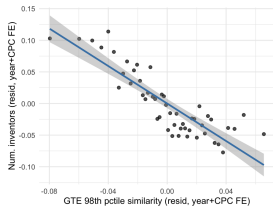
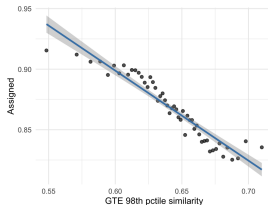
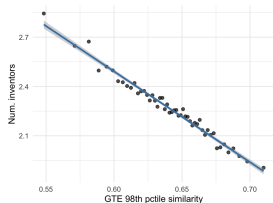
Component	General	Baseline ($\alpha = 1, \eta = 1$)
Entry cost	$f(H) = \phi H^\alpha$	ϕH
R&D cost	$c(q) = \frac{1}{2}\gamma q^{1+\eta}$	$\frac{1}{2}\gamma q^2$
Spacing growth	$g_d = \frac{\alpha}{2}g_H$	$\frac{1}{2}g_H$
Quality growth	$g_q = g_d$	$\frac{1}{2}g_H$
Entry growth	$g_n = (1 - \frac{\alpha}{2})g_H$	$\frac{1}{2}g_H$
Fixed cost growth	$g_f = \alpha g_H$	g_H

R&D growth equation:

$$g_{R\&D} = \underbrace{(1 - \frac{\alpha}{2})g_H}_{\text{Entry}} + \underbrace{\frac{\theta\alpha}{2}g_H}_{\text{Quality scaling}} + \underbrace{\frac{\theta\eta\alpha}{2}g_H}_{\text{Fishing out}} + \underbrace{(1 - \theta)\alpha g_H}_{\text{Burden of knowledge}}$$

Backup: Spacing Predicts Investment and Quality

◀ Return



Model: $q^* = d/\gamma$ — more isolated inventors invest more in quality

4 q proxies: Co-inventors, assigned patents, 5-yr citations, KPSS patent value

Confirmed in cross-section and within technology classes over time

▶ Regression results

Spacing and Quality: Regression Results

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	Co-Inventors	Firm Assignment	Citations (5-yr)	KPSS Value
Panel A: Cross-sectional, mean similarity				
GTE	−2.545*** (0.679)	−0.659*** (0.180)	6.347 (4.480)	−22.093** (9.036)
PaECTER	−4.928*** (1.002)	−1.723*** (0.424)	−36.681*** (8.383)	−122.904*** (46.101)
Panel B: Cross-sectional, 98th percentile similarity				
GTE	−1.477** (0.670)	−0.253* (0.131)	4.788 (3.220)	−32.531*** (7.022)
PaECTER	−1.067 (1.592)	−1.174*** (0.431)	−19.404*** (6.443)	−84.301* (43.556)
Observations	219,772	219,772	219,772	3,274,108
Fixed Effects	Year + CPC Class			
Panel C: Time-series, Δ mean similarity, CUSP (1836–2023)				
GTE	−0.011* (0.005)	—	—	—
PaECTER	−0.004** (0.001)	—	—	—
Observations	15,813			
Fixed Effects	Year + CPC Class			

Backup: Validation of Spillover Elasticity

◀ Return

Quasi-experimental estimates of R&D spillovers Bloom et al. 2013, Lucking et al. 2019

- Firm-level TFP elasticity to shocks to spillover pool
- SPILLTECH $\equiv$ sum of neighbors' R&D, weighted by idea distance
- IV: State R&D tax credit shocks

Ingredients:

- Bloom et al. 2013 (1981-2001): $\hat{\beta} = \mathbf{0.206}$
- Standard deviation $\log(\text{SPILLTECH}) = \mathbf{1.04}$
- Avg change in similarity (Bloom sample period): $\mathbf{-0.007\sigma/\text{yr}}$

Implied TFP drag from average rate of spreading out:

- $-0.007 \times 1.04 \times 0.206 = \mathbf{-0.15\%/yr}$ ✓

Backup: Forces Reducing Research Productivity

[◀ Return](#)

Forces reducing g_{TFP} :

$$g_{TFP} = \frac{dA}{dH} \cdot \dot{H} = \left(\underbrace{\frac{dq}{dH} \left[1 + \beta \left(1 - \frac{d}{\lambda} \right) \right]}_{\text{(1) Spillover attenuation: mult.}\downarrow} - \underbrace{\frac{\beta q}{\lambda} \frac{dd}{dH}}_{\text{(2) Spillover base: } q\uparrow} - \underbrace{\frac{\tau}{4} \frac{dd}{dH}}_{\text{Adapt. drag}} \right) \cdot \underbrace{\left[g_H^* + \frac{\delta \beta}{\gamma} \left(1 - \frac{d}{\lambda} \right) \right]}_{\text{(3) Frontier spillovers}\downarrow \text{ as } d\uparrow} \cdot H$$

Forces increasing Agg R&D:

$$\frac{d(\text{Agg R\&D})}{dH} = \underbrace{\frac{dn}{dH} \cdot [c(q) + f(H)]}_{\text{(6) Entry expansion}} + \underbrace{n \cdot c'(q)}_{\text{(4) Fishing out}} \cdot \underbrace{\frac{dq}{dH}}_{\text{(7) Quality scaling}} + \underbrace{n \cdot f'(H)}_{\text{(5) Burden of knowledge}}$$

Spatial Forces: 1, 2, 3, 6, 7 — **Traditional: 4, 5**

Backup: TFP Estimation Strategy

1. Theory

$$g_{TFP} = \underbrace{\frac{dq}{dt} \left[1 + \beta \left(1 - \frac{d}{\lambda} \right) \right]}_{\text{local ideas + spillovers}} - \underbrace{\left(\frac{\beta q}{\lambda} + \frac{\tau}{4} \right) \frac{dd}{dt}}_{\text{spreading losses}}$$

[► Derivation](#)[◀ Return](#)

With $q^* = d/\gamma$:

$$g_{TFP} = \underbrace{\kappa_1}_{\text{net quality effect}} \frac{dd}{dt} - \underbrace{\kappa_2}_{>0} d \frac{dd}{dt}$$

2. Regression

[► Table](#)

$$\Delta \log(\text{TFP})_t = b_0 + b_1(-\Delta \text{Sim}_t) + b_2(-\Delta \text{Sim}_t)(-\text{Sim}_{t-1}) + b_3 t + \epsilon_t$$

b_1 : quality gains net of adaptation costs

$b_2 < 0$: spreading weakens spillovers more at greater distance

3. Quantification

At observed spreading and
1991 spacing

$$\widehat{\text{drag}} = -0.16 \text{ pp/yr}$$

Explains **7%** of the
1948–2015 TFP-growth
slowdown

Backup: R&D Estimation Strategy

1. Aggregate R&D accounting

[► Derivation](#)[◀ Return](#)

$$g_{R\&D} = \underbrace{g_n}_{\text{inventions}} + \underbrace{\theta(1+\eta)g_q}_{\text{variable quality costs}} + \underbrace{(1-\theta)g_f}_{\text{fixed costs}}$$

2. Equilibrium relationships

$$n = H/d, \quad q = d/\gamma, \quad f(H) \propto H^\alpha$$

$$\Downarrow$$

$$g_n = g_H - g_d, \quad g_q = g_d, \quad g_f = \alpha g_H$$

3. Substitute into aggregate R&D growth

$$g_{R\&D} = \underbrace{[1 + \alpha(1-\theta)]g_H}_{a_0} + \underbrace{[\theta(1+\eta)-1]g_d}_{a_1 g_d}$$

Empirical equation ($g_d \approx -\Delta \text{Sim}_t$): $g_{R\&D,t} = a_0 + a_1(-\Delta \text{Sim}_t) + \epsilon_t$

[► Regression table](#)

Slope identifies the variable-cost share

$$a_1 = \theta(1+\eta) - 1 \quad \Rightarrow \quad \boxed{\theta = \frac{a_1 + 1}{1 + \eta}}$$

Intercept identifies frontier growth

$$a_0 = [1 + \alpha(1-\theta)]g_H \quad \Rightarrow \quad \boxed{g_H = \frac{a_0}{1 + \alpha(1-\theta)}}$$

Backup: Calibrating the Model

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What we assume (baseline):

Parameter	Value	Source/Interpretation
θ (variable cost share)	0.69	NSF survey; regression estimate (0.72)
g_H (idea space growth)	2.7%/yr	Two methods: $4.0\% = g_{R\&D} = 3/2 \cdot g_H$; R&D regression constant a_0 ($\alpha = 1$)
α (entry cost curvature)	1.0	Entry costs scale linearly with H
η (R&D cost curvature)	1.0	Quadratic R&D costs

Backup: Detailed Spatial Decomposition

This is a **model-conditioned decomposition**, disciplined by external checks above.

Calibration: $\theta = 0.69$ (NSF) $g_H = 2.7\%/yr$ (Model identity $g_{R\&D} = \frac{3}{2}g_H$) $\alpha = 1, \eta = 1$ (baseline)

Backup: Detailed Spatial Decomposition

This is a **model-conditioned decomposition**, disciplined by external checks above.

Calibration: $\theta = 0.69$ (NSF) $g_H = 2.7\%/yr$ (Model identity $g_{R\&D} = \frac{3}{2}g_H$) $\alpha = 1, \eta = 1$ (baseline)

Component	Contribution
<i>TFP deceleration</i>	-1.6%/yr
Spatial drag worsened	-0.11%/yr
Unmodeled factors	-1.49%/yr
<i>R&D growth</i>	+4.0%/yr
Total R&D productivity decline, 1948–2015	-5.6%/yr

Spatial drag: Knowledge spillovers fade

Backup: Detailed Spatial Decomposition

This is a **model-conditioned decomposition**, disciplined by external checks above.

Calibration: $\theta = 0.69$ (NSF) $g_H = 2.7\%/yr$ (Model identity $g_{R\&D} = \frac{3}{2}g_H$) $\alpha = 1, \eta = 1$ (baseline)

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Spatial drag worsened	-0.11%/yr
Unmodeled factors	-1.49%/yr
<i>R&D growth</i>	+4.0%/yr
Entry expansion $(1 - \frac{\alpha}{2})g_H$	+1.33%/yr
Quality scaling $(\theta \eta \frac{\alpha}{2}g_H)$	+0.92%/yr
Fishing out $(\theta \eta \frac{\alpha}{2}g_H)$	+0.92%/yr
Burden of knowledge $(1 - \theta)(\alpha g_H)$	+0.83%/yr
Total R&D productivity decline, 1948–2015	-5.6%/yr

Entry expansion: More inventions on more territory

Quality scaling: Each inventor builds bigger, raising cost per unit TFP

Backup: Detailed Spatial Decomposition

This is a **model-conditioned decomposition**, disciplined by external checks above.

Calibration: $\theta = 0.69$ (NSF) $g_H = 2.7\%/yr$ (Model identity $g_{R\&D} = \frac{3}{2}g_H$) $\alpha = 1, \eta = 1$ (baseline)

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Fishing out $(\theta \eta \frac{\alpha}{2}g_H)$	+0.92%/yr
Burden of knowledge $(1 - \theta)(\alpha g_H)$	+0.83%/yr
Total R&D productivity decline, 1948–2015	-5.6%/yr
Spatial contribution (42%)	-2.36%/yr
Non-spatial contribution (58%)	-3.24%/yr

Spatial forces account for
42% of the long-run
productivity decline

Sensitivity: **credible range 42–58%**; baseline (42%) is conservative; better-identified calibrations higher

Backup: TFP Regression Results

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	Annual		3 - Year	5 - Year
$b_1 : -1 \times \Delta \text{Sim}$	-0.169*** (0.057)	-0.171*** (0.083)	-0.278*** (0.095)	-0.269*** (0.098)
$b_2 : (-1 \times \Delta \text{Sim}) \times (-1 \times \text{Sim}_{t-1})$	—	-0.015 (0.342)	-0.408 (0.320)	-0.571* (0.312)
<i>Implied TFP drag from spreading out ($\overline{\Delta \text{Sim}}$, %/yr):</i>				
1948 (Sim = 0.35)	-0.08	-0.08	-0.07	-0.04
1991 (Sim = 0, baseline)	-0.08	-0.09	-0.14	-0.16

$\hat{b}_1 < 0$: adaptation drag > quality investment $\hat{b}_2 < 0$: spillover attenuation confirmed ✓

Data: TFP and Real R&D, 1948–2015 (Bloom et al., 2020)

Backup: Implied $\theta = 0.72$ and $g_H = 2.7\%/yr$

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	Annual	3 - Year	5 - Year
$\alpha_1: -1 \times \Delta Sim$	0.165 (0.177)	0.448** (0.219)	0.438* (0.244)
α_0 : Constant	0.034*** (0.006)	0.102*** (0.013)	0.173*** (0.018)
Implied θ (variable cost share)	0.58	0.72	0.72
Implied g_H (idea space growth)	2.4%/yr	2.7%/yr	2.7%/yr

- $\theta = 72\%$ aligns with NSF survey data (labor = 69% of R&D) ✓
- $g_H = 2.7\%/yr$ consistent w/ embedding volume growth & $g_{R\&D} = 3/2 \cdot g_H$ ✓ 
- Model predicts $g_d/g_{R\&D} = 1/3$; in data, $-\Delta Sim/g_{R\&D} \approx 0.3$ ✓

Backup: Spatial Share Higher for Better-Identified Calibrations

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Dominant uncertainty: α (entry cost curvature)

Evidence pins $\alpha \approx 0.67$ unique inventor growth (2.0%/yr)

$\eta = 0.625$ from quasi-experiment Guceri-Liu 2016

Baseline is conservative: Credible range 42–58%

α	Spatial share (%)	
	$\eta = 1$	$\eta = 0.625$
0.50	55	58
0.75	48	51
1.00	42	46
1.25	37	41
1.50	33	37